Every Timeline Is an Opinion

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Abstract

Where does interpretation come from? Computation requires an interpreter. A symbol means whatever the hardware does when it receives that symbol. But hardware is physics, and we see physics through the lens of our own abstractions. This regress has led some to posit a primitive interpreter that grounds computation in experience. I argue the interpreter is not a primitive. Change is the primitive. Each change is contentless. It is a state defined only by its difference from the other changes. An aspect of reality is a set of changes. A timeline is a sequence of changes. As a timeline progresses, it preserves some aspects of reality and destroys others. That is already a value judgement, before any organism exists. Every timeline is an opinion about what ought to persist. Representation collapses into value judgement at the most basic ontological level. I formalise this using Stack Theory, a framework in which everything is a potentially infinite stack of abstraction layers, each interpreted by the layer below. I show that this ontological natural selection implies self-producing living systems rather than presupposing them. Once such systems exist, the Psychophysical Principle of Causality governs how they partition the world into objects and properties. Under generalisation-optimal learning, the only representational primitive that does not raise a floor on free energy is valence. Objects and properties are not neutral categories. They are patterns in attraction and repulsion. Autopoiesis is usually assumed. Here I show it is implied by the very fact of change.

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Introduction

Artificial life studies what life could be (Langton, 1995; Bedau, 2003). I want to know why it must be. Living things interpret their world in order to survive. The world interprets the actions of the living and decides who survives. But where does interpretation come from?

Computation requires interpretation. A word of machine code triggers whatever process the hardware is designed to perform when it receives that word. Change the hardware, change the meaning. Software is a state of hardware, not a separate substance. I named the contrary error "computational dualism" (Bennett, 2024a). The mechanistic account of computation (Piccinini, 2015) treats computational steps and storage as features of the mechanism itself. I argue they are features of the vocabulary used to describe it (Bennett, 2026a). Turing computation is not intrinsic to physics. It depends on an external interpreter that partitions continuous dynamics into discrete meaningful states.

A human made a Turing machine. A human interpreted the world to create an interpreter. But where does the interpreter before that come from?

This is not just a question for philosophy of mind. It is the same question artificial life asks about the origins of living systems. A living system is one that maintains itself. It interprets its environment and acts to preserve its own existence. But if interpretation requires a pre-existing interpreter, we have a chicken-and-egg problem. Autopoiesis is typically assumed as a starting point (Maturana and Varela, 1980; Varela et al., 1974). I want to derive it.

If the interpreter is treated as primitive, the question is just pushed back one step. Infinite regress. In prior work I showed how we can avoid treating interpreters as primitive. Every interpreter is just another layer in the stack, constituted by the physical dynamics of the layer below (Bennett, 2024a, 2025a). Here I go further. I argue the interpreter is the timeline itself. A timeline is a sequence of states, each marking a point of difference. As it progresses, it preserves some aspects of reality and destroys others. That preservation and destruction is already a value judgement. Every timeline is an opinion about what ought to persist.

I formalise this using Stack Theory (Bennett, 2025a, 2024a,b), a framework in which everything is a potentially infinite stack of abstraction layers. The novel claim of this paper is that the interpreter problem from the computation and consciousness literature is the same problem as the origins of life. Both ask where interpretation comes from. Both are answered by the timeline. Normativity is not injected from outside, but built by the structure of change itself. Self-producing systems are a consequence, not a premise. An infinite set of timelines is an infinite set of normative physics.

The Minimal Ontology

I want to reason about what must exist in any conceivable environment. Not what exists in our particular universe, but what holds regardless of which stack we inhabit.

We cannot know what is "really there" at the bottom of our stack, because we see everything through the lens of our own abstractions. Derrida's *différance* applies (Derrida, 1978). We get an infinite regress. Instead of trying to get physics from consciousness (the IIT approach (Tononi, 2004)), I try to get to something like consciousness from physics. But I cannot assume any particular physics. I can only assume what must hold generally.

The answer is change. Without change or difference, everything would be the same thing. If everything is the same, there is nothing. Difference is foundational to existence. Schrödinger asked how life maintains its organisation against entropy (Schrödinger, 1944). I start further back. Before asking how life persists, I ask why anything persists at all. This is close in spirit to Whitehead's process philosophy, where becoming is more fundamental than being (Whitehead, 1929). It also connects to Prigogine's account of time as irreducible, where dissipative structures persist far from equilibrium through continuous exchange with their environment (Prigogine, 1980). Dennett argued that which patterns count as real depends on the vocabulary used to describe them (Dennett, 1991). Ladyman and Ross generalised this into a structural realism in which no description level is metaphysically privileged (Ladyman and Ross, 2007). From this minimal assumption I adopt the following definitions (Bennett, 2025b,a).

Definition 1 (Environment, programs, and vocabularies)**.** *An* environment *is a nonempty set Φ of mutually exclusive states. A* declarative program *is any set of states $p \subseteq \Phi$. Let $\mathcal{P} := 2^{\Phi}$ denote the set of all programs. An* aspect *of the environment is a set x of programs such that $\bigcap_{p\in x} p \neq \emptyset$. A* vocabulary *is any set of programs $\mathfrak{v} \subseteq \mathcal{P}$. Unless specified otherwise, vocabularies are finite.*

Each state marks a point of difference. I assign states no specific ontological content. No semantics, no internal structure. Just that there is more than one of them, and that they differ. A program is a constraint that holds in some states and not others. A vocabulary represents what a bounded system can express. Finiteness of vocabularies follows from spatial extension. In our universe, the Bekenstein bound enforces this (Bekenstein, 1981). In any conceivable universe, a spatially bounded system can only instantiate finitely many distinctions. Landauer showed that computation is physical and that information erasure costs energy (Landauer, 1961). The formalism takes this seriously. There are no disembodied symbols here.

Definition 2 (Abstraction layer and statements)**.** *A vocabulary $\mathfrak{v}$ induces an embodied language*

$$L_{\mathfrak{v}} = \{l \subseteq \mathfrak{v} \mid \bigcap_{p\in l} p \neq \emptyset\}.$$

*Elements $l \in L_{\mathfrak{v}}$ are called **statements**. The **truth set** of l is $T(l) := \bigcap_{p\in l} p \subseteq \Phi$. A statement is **true** at state ϕ iff $\phi \in T(l)$.*

A statement is a physical configuration that a system can be in. When I raise my arm, my body makes a statement. The truth set is the set of states in which that configuration holds. By convention, the intersection of the empty family is Φ, so $\emptyset \in L_{\mathfrak{v}}$ and is true at every state. An abstraction layer is a window onto part of the environment, not a simplified map of something else.

Definition 3 (Extensions and equivalence)**.** *For any set of programs $x \subseteq \mathfrak{v}$, a **completion** of x is any $y \in L_{\mathfrak{v}}$ with $x \subseteq y$. The **extension** of x is $\mathrm{Ext}(x) = \{y \in L_{\mathfrak{v}} \mid x \subseteq y\}$. I say x and y are **equivalent** iff $\mathrm{Ext}(x) = \mathrm{Ext}(y)$.*

Completions add detail to a coarse description. The extension of a statement contains all the ways it could be refined. This is a formal counterpart of Kauffman's adjacent possible (Kauffman, 2000). Weakness, generalisation, and the stack bottleneck results are all functions of extensions (Bennett, 2024b,a).

Timelines as Opinions

Definition 4 (Objective time and trajectories)**.** *Objective time is indexed by $u \in \mathbb{N}$. An **objective trajectory** is a function $\tau : \mathbb{N} \to \Phi$ such that $\tau(u+1) \neq \tau(u)$ for all u.*

A trajectory is one possible history of the environment. A possible world. The condition $\tau(u+1) \neq \tau(u)$ means every tick is a genuine transition. Time is just the passage of change. As a timeline progresses, some statements remain true and others cease to be. An atom persists because it is compatible with the physics of that timeline. A sand castle does not, because wind and rain destroy it. Persistence is survival. Darwin's natural selection, applied to every aspect of the environment, not just living things (Bennett, 2025a). Campbell called this "blind variation and selective retention" and argued it operates across knowledge processes (Campbell, 1960). Dawkins coined the term "universal Darwinism" for the claim that natural selection applies wherever there is variation, heredity, and differential fitness (Dawkins, 1983). Smolin applied natural selection to universes themselves (Smolin, 1997). Godfrey-Smith gave the most careful philosophical analysis of what makes something a Darwinian population (Godfrey-Smith, 2009). Okasha formalised how selection operates at multiple levels simultaneously (Okasha, 2006). My cosmic ought is more radical than any of these. It applies selection to existence itself, before replication, before populations, before any particular physics.

Definition 5 (Persistence)**.** *A statement $l \in L_{\mathfrak{v}}$ **persists** along a trajectory τ over an interval $[u_1, u_2]$ if for every $u \in [u_1, u_2]$, l is true at $\tau(u)$.*

This is the foundation of normativity. As a timeline transitions from one state to another, it eliminates that which does not preserve itself. This is not a metaphor. It is a structural fact about sequences of mutually exclusive states. Some aspects survive the transition. Others do not. The timeline does not "choose" in any intentional sense. But what persists is what the timeline "values".

Proposition 1 (Timelines are opinions)**.** *Let τ and τ' be distinct trajectories on the same environment Φ, and fix a vocabulary $\mathfrak{v}$. For any interval I, define the persistence set $\mathcal{S}_{\tau,I} := \{l \in L_{\mathfrak{v}} \mid l \text{ persists along } \tau \text{ over } I\}$. If there exist $u^* \in \mathbb{N}$ and a program $p \in \mathfrak{v}$ such that $\tau(u^*) \in p$ and $\tau'(u^*) \notin p$, then $\mathcal{S}_{\tau,[u^*,u^*]} \neq \mathcal{S}_{\tau',[u^*,u^*]}$.*

Proof. The singleton $\{p\}$ lies in $L_{\mathfrak{v}}$ because $\tau(u^*) \in p$, so $\bigcap_{q\in\{p\}} q = p \neq \emptyset$. It persists along τ over $[u^*, u^*]$ because $\tau(u^*) \in p$. It does not persist along τ' over $[u^*, u^*]$ because $\tau'(u^*) \notin p$. So $\{p\} \in \mathcal{S}_{\tau,[u^*,u^*]}$ but $\{p\} \notin \mathcal{S}_{\tau',[u^*,u^*]}$, and the persistence sets differ. □

Trajectories disagree about what persists. But they also disagree about how much things persist. That disagreement is a normative ordering.

Definition 6 (Persistence ordering)**.** *Fix a vocabulary $\mathfrak{v}$, a trajectory τ, and an interval $[u_1, u_2]$. For $l \in L_{\mathfrak{v}}$, define the **persistence measure***

$$\rho_{\tau,[u_1,u_2]}(l) := |\{u \in \{u_1, \dots, u_2\} \mid l \text{ is true at } \tau(u)\}|.$$

Write $l \geq_{\tau,[u_1,u_2]} l'$ iff $\rho_{\tau,[u_1,u_2]}(l) \geq \rho_{\tau,[u_1,u_2]}(l')$.

The persistence measure counts how many time steps in an interval a statement survives. It induces a total preorder on $L_{\mathfrak{v}}$. This is what the timeline values. Statements that survive more time steps rank higher.

Proposition 2 (Timelines rank differently)**.** *Under the conditions of Proposition 1, the persistence orderings $\geq_{\tau,[u_1,u_2]}$ and $\geq_{\tau',[u_1,u_2]}$ can differ. There exist an interval and statements l, l' such that $l \geq_{\tau,[u_1,u_2]} l'$ but $l \not\geq_{\tau',[u_1,u_2]} l'$.*

Proof. Take u^* and p as in Proposition 1, with $\tau(u^*) \in p$ and $\tau'(u^*) \notin p$. Set $l = \{p\}$ and $l' = \emptyset$. Consider the singleton interval $[u^*, u^*]$. The empty statement is true at every state, so $\rho_{\tau,[u^*,u^*]}(l') = \rho_{\tau',[u^*,u^*]}(l') = 1$. Since $\tau(u^*) \in p$, we have $\rho_{\tau,[u^*,u^*]}(l) = 1$. Since $\tau'(u^*) \notin p$, we have $\rho_{\tau',[u^*,u^*]}(l) = 0$. So $l \geq_{\tau,[u^*,u^*]} l'$ but $l \not\geq_{\tau',[u^*,u^*]} l'$. □

Every timeline is an opinion. This is now a formal statement, not an opinion. Each trajectory induces a total preorder over statements. Different trajectories induce different preorders. What one timeline ranks as equally persistent, another ranks as less persistent. The normative ordering arises from the structure of change alone. No agent imposed it. No organism chose it. The timeline decided.

At any abstraction layer rich enough to distinguish them, different timelines express different "oughts". The set of all possible timelines is the set of all possible opinions about what ought to exist. A particular timeline determines a particular physics. It is the interpreter. It determines the arbitrary associations between what exists now and what exists next. This is not just renaming persistence as value. The renaming does work. It connects persistence to weakness maximisation, weakness to self-production, and self-production to valence-first representation. The chain from timeline to living system runs through that connection.

In computational dualism, the interpreter partitions continuous physics into discrete meaningful symbols (Bennett, 2024a). A timeline does the same thing, but through selection rather than choice. What persists is what the timeline distinguishes. What does not persist is noise from that timeline's perspective. The partitioning is not performed by an external agent. It is performed by the passage of time itself. Moreover, different vocabularies watching the same physical process can disagree about whether anything has happened, because layer time advances only when the encoding changes (Bennett, 2026a). The timeline's opinion is not just about what persists. It is about what counts as change.

States are contentless not because they lack content, but because any content we ascribe is itself just another aspect defined by relations between states (Bennett, 2025a). This is why the regress does not arise. The base states are not abstracted from anything. They just are.

An objection arises. Everything above is vocabulary-relative. The persistence ordering depends on which vocabulary you use. Different vocabularies can disagree about whether anything has changed at all (Bennett, 2026a). So has the interpreter really been eliminated, or just relocated into the choice of vocabulary? The answer is that the vocabulary IS the embodied system. A bacterium and a human watch the same physics through different vocabularies and arrive at different persistence orderings. There is no vocabulary-independent fact of the matter, and that is not a defect. It is the point. There is no view from nowhere. But the structure of the argument holds in every vocabulary. Change implies persistence, persistence implies an ordering, and the ordering is always there regardless of which vocabulary expresses it. What changes across vocabularies is which statements rank where. What does not change is that there is a ranking.

From Ought to Life

The cosmic ought establishes that persistence is a value judgment made by the timeline. But why should self-producing systems emerge from this?

Persistence favours weakness

A policy is a statement interpreted as a constraint on behaviour. The weakness of a policy π is $|\mathrm{Ext}(\pi)|$, the number of further commitments that remain available while staying correct (Bennett, 2024b). A weaker policy is compatible with more situations and generalises better. The Price equation (Price, 1970) gives the most abstract formulation of selection as change in a population mean. Here selection operates on something even more basic than populations. It operates on what exists.

Persistence along a timeline biases surviving statements toward weakness. A weaker statement leaves more future refinements available. Under a maximally uninformative model over extensions, this increases the probability of persistence under novelty (Bennett, 2026b). This proves a formal version of the law of increasing functional information proposed by Wong et al. (Wong et al., 2023). It formalises the connection between persistence (temporal) and weakness (combinatorial). Time and space are formally the same underlying resource at a finite abstraction layer. Temporal accumulation of distinctions is exactly spatial distinction in a lifted vocabulary, and elapsed time is exactly a space count in a tick vocabulary (Bennett, 2026a).

This is not to say that complexity must increase. I mean only that selection under novelty biases survivors toward weak constraints that preserve compatibility with unseen outcomes (Bennett, 2026b). The free energy proxy F_2 in the theorem below is a discrete analogue of variational free energy (Friston, 2010). Kirchhoff et al. connected autopoiesis to the free energy principle through Markov blankets (Kirchhoff et al., 2018). The present account arrives at a similar destination from different premises.

Why should self-production be favoured? There are physics-based answers. Dissipative structures persist far from equilibrium because they maximise entropy production. Self-replicators are thermodynamically favoured under driven conditions. Autocatalytic cycles can sustain self-reproduction in chemical systems. All of these assume a particular physics or chemistry (England, 2013; Schneider and Kay, 1994; Eigen and Schuster, 1979). I don't need to assume any particular physics or chemistry. After all, that would be assuming an interpreter. Computational dualism again. No, I'm just pointing out that a system that can regenerate the conditions for its own persistence has a larger viable continuation set than one that cannot, because it can recover from perturbations that would destroy a non-regenerating system. Self-production should therefore enlarge the viable continuation set.

Consider two configurations in a stable environment. A rigid crystal has few programs in its policy. If any one is disrupted, the crystal is destroyed. Its extension is small because survival requires every component to hold. A self-repairing membrane has more programs, but its repair mechanism means it can recover from disruptions that would destroy the crystal. The membrane's policy is compatible with more future states. It is more weakly constrained. The crystal is simpler but the membrane is weaker, and weakness is what persistence under novelty selects for. This is why self-producing organisation is favoured despite its complexity. The full derivation, including the three orders of persistence (static, dynamic, and novelty-generating) and the formal conditions under which weakness and simplicity diverge, is given by the autopoietic theorem (Bennett and Suzuki, 2026). This paper provides the ontological premises from which that derivation begins. It is complementary.

Persistence biases toward self-producing organisation. This is autopoiesis (Maturana and Varela, 1980), derived from persistence and weakness rather than assumed.

Representation collapses into value judgment

Once self-preserving systems exist, how should they partition the world into objects and properties? One option is to assume objects and learn causal relations between them. The other is to assume valence is given and learn the objects that fit it (Bennett, 2023, 2025a; Bennett et al., 2024). Valence is the attraction to or repulsion from states. A living system interpreting things as good and bad. Since we already have normativity from The Cosmic Ought, we get valence for free. It is the viability signal, the most basic evaluation a system can make. The Psychophysical Principle of Causality (Bennett, 2025a; Bennett et al., 2024) proves the second option is the only one consistent with generalisation-optimal learning.

Theorem 1 (The Psychophysical Principle of Causality (Bennett, 2025a))**.** *Let μ be a viability statement encoding the most basic valence partition (viable versus not viable). For any set of programs $x \subseteq \mathfrak{v}$, define the viable continuation set $\Omega_x := \mathrm{Ext}(x) \cap \mathrm{Ext}(\mu)$. Define the free energy proxy*

$$F_2(x) := \log_2 |\mathrm{Ext}(\mu)| - \log_2 |\Omega_x|.$$

Let $c \in L_{\mathfrak{v}}$ be a commitment not forced by viability, meaning there exists $o \in \mathrm{Ext}(\mu)$ with $c \not\subseteq o$. Assume Ω_x contains at least one viable continuation that violates c. Then either $\Omega_{x\cup c} = \emptyset$ or $F_2(x \cup c) > F_2(x)$.

Proof. Since $x \cup c \supseteq x$, any $y \in L_{\mathfrak{v}}$ with $x \cup c \subseteq y$ also satisfies $x \subseteq y$. So $\mathrm{Ext}(x \cup c) \subseteq \mathrm{Ext}(x)$. Intersecting both sides with $\mathrm{Ext}(\mu)$ gives $\Omega_{x\cup c} \subseteq \Omega_x$. By hypothesis there exists $o \in \Omega_x$ with $c \not\subseteq o$. Since $c \not\subseteq o$, we have $x \cup c \not\subseteq o$, so $o \notin \mathrm{Ext}(x \cup c)$ and hence $o \notin \Omega_{x\cup c}$. The inclusion is therefore strict. If $\Omega_{x\cup c} = \emptyset$, the first case holds. Otherwise $0 < |\Omega_{x\cup c}| < |\Omega_x|$. Since $\log_2$ is strictly increasing, $\log_2 |\Omega_{x\cup c}| < \log_2 |\Omega_x|$, and so $F_2(x\cup c) = \log_2 |\mathrm{Ext}(\mu)| - \log_2 |\Omega_{x\cup c}| > \log_2 |\mathrm{Ext}(\mu)| - \log_2 |\Omega_x| = F_2(x)$. □

Adding an unnecessary representational commitment strictly raises free energy whenever it rules out even one viable future. The only safe primitive is valence. Everything else must be learned as a revisable causal identity (Bennett, 2023). Objects and properties are not neutral categories with valence attached after the fact. They are patterns in attraction and repulsion. A dog means something very different to a flea than it does to a human, because the causal identities they construct are shaped by different survival pressures (Bennett, 2025a). Evaluation precedes representation. This is the Psychophysical Principle of Causality in one sentence.

Though the Cosmic Ought features prominently in my thesis, in related papers on consciousness I begin with embodiment Bennett (2023); Bennett et al. (2024). Here, I draw the line directly to The Cosmic Ought. This connects time and existence to the enactivist tradition (Varela et al., 2016; Thompson, 2007; Paolo, 2005; Ciaunica et al., 2023), which holds that cognition is not the recovery of a pre-given world but the enactment of a world through embodied interaction. Weber and Varela argued that normativity arises from autopoiesis, that life after Kant grounds purposes in self-production (Weber and Varela, 2002). Barandiaran et al. derived agency from the same foundation (Barandiaran et al., 2009). Lyon extended cognition all the way down to bacteria (Lyon, 2006). Hoffmeyer argued that organisms are fundamentally sign-users, partitioning their world through biosemiotic processes (Hoffmeyer, 1996). Developmental evidence suggests that co-embodied, co-homeostatic interaction shapes cognition from the earliest stages of life (Ciaunica et al., 2021). I agree with all of this, but I locate the foundation one level deeper. Normativity does not begin with autopoiesis. It begins with the timeline.

Discussion

The interpreter problem and the origins of life problem are the same problem. Both ask where interpretation comes from. The answer is the timeline. A timeline preserves some aspects and destroys others. That is already interpretation.

This paper complements two related results. The law of increasing functional information shows that under novelty selection, persistence biases survivors toward weak constraints (Bennett, 2026b, 2025a). That is what happens once the cosmic ought is in place. The autopoietic theorem (Bennett and Suzuki, 2026) formalises organisational closure in Stack Theory terms. That is what self-producing systems do once they exist. This paper provides the ontological foundation for both.

Related work

Jonas argued in *The Phenomenon of Life* that normativity first arises with metabolism (Jonas, 1966). The moment a system must actively maintain itself against entropy, it has a stake in its own existence. I agree that metabolism is where normativity becomes visible to the organism. But I argue normativity exists before metabolism. The timeline preserves and destroys before any organism exists. Jonas locates the first freedom in the cell. I locate it in change itself. His account explains why organisms care. Mine explains why there is something to care about.

Pattee's "epistemic cut" (Pattee, 2001) identifies the same structural problem I call computational dualism. He argues that symbolic information and physical dynamics are irreducibly complementary, and that this complementarity first arises with self-replication (Pattee, 1995). I agree with the diagnosis but not the conclusion. Pattee treats the epistemic cut as irreducible. I dissolve it. In Stack Theory, the interpreter is just another layer constituted by contentless states at the layer below. Representation collapses into value judgment via the Psychophysical Principle. There is no irreducible cut. There is just the stack. Independent convergence on the same diagnosis has recently appeared in the philosophical literature under the name "Abstraction Fallacy" (Lerchner, 2026). That someone else reached the same conclusion via a different path lends support.

Rosen's "closure to efficient causation" (Rosen, 1991) and the subsequent formal account of closure of constraints (Montévil and Mossio, 2015; Moreno and Mossio, 2015) characterise living systems as those in which every component required for maintenance is produced within the system. This is precise and influential, but it characterises what living systems look like without explaining why they emerge. I show self-producing systems are implied by the cosmic ought rather than assuming them. Ruiz-Mirazo et al. define life as a conjunction of autonomy and open-ended evolution (Ruiz-Mirazo et al., 2004). I derive the conditions for both from weaker premises.

Deacon's "Incomplete Nature" (Deacon, 2012) argues that constraints rather than forces produce teleological phenomena. His "autogen" is a self-producing system derived from thermodynamic reasoning. The spirit is close to mine. The difference is that his argument is philosophical whereas mine is formal. I prove that persistence under change biases toward weakness, and that representation must collapse into value judgment.

Dennett's "Darwin's Dangerous Idea" (Dennett, 1995) extends natural selection as a universal acid that dissolves traditional explanatory boundaries. My cosmic ought is a more radical version. Dennett applies selection to biology and culture. I apply it to existence itself, before organisms, before chemistry, before any particular physics. The timeline is the first selector. Assembly Theory (Sharma et al., 2023) offers a related combinatorial account of selection and complexity but does not derive normativity from first principles. Kauffman's work on self-organisation and autonomous agents (Kauffman, 1993, 2000) identifies the adjacent possible as a driver of biological novelty. In my formalism, the adjacent possible is the extension of a policy. What Kauffman frames as an empirical tendency, I formalise as a selection pressure.

Solé et al. identify fundamental constraints on the logic of living systems (Solé et al., 2024). Walker et al. argue that life requires a specific informational architecture (Walker et al., 2016). Levin's technological approach to mind everywhere treats biological systems as multiscale competency architectures, where intelligence operates at every level from cells to organisms (Levin, 2022). The stack formalises this hierarchy. These characterise necessary conditions for life, but not why living systems emerge. Any environment with change already has a cosmic ought, and self-producing systems are what persistence under that ought selects for. The enactive AI programme (Froese and Ziemke, 2009) seeks to build systems that enact their own worlds through embodied interaction. The argument of this paper suggests why that programme is on the right track.

Limitations

The cosmic ought gives the direction of selection, not its timescale. It does not predict when or where life arises, only that persistence under change favours self-producing systems over rigid ones. The mechanism of origination is empirical. What is derived is that the direction of selection points toward autopoiesis whenever the environment is stable enough for weakness and simplicity to diverge. The formal result in Bennett (2026b) assumes exchangeable novelty, meaning the prior over future demands depends only on their number, not on which demands they are. The companion paper's supplementary material bounds how far novelty must depart from exchangeability before the weakness ordering reverses (Bennett, 2026b). Within that bound, the argument holds.

Implications for artificial life

The origins of life are not a problem of chemistry or physics in the narrow sense. They are a problem of selection under change. Any environment that has states and transitions already has a cosmic ought. Self-producing systems emerge when this selection pressure iterates across scales, because systems that regenerate the conditions for their own persistence are more weakly constrained and thus more likely to survive novel transitions. Open-ended evolution, the central unsolved problem of artificial life (Packard et al., 2019), requires exactly this iterative escalation.

This resolves a tension between computation and life. If computation requires a pre-existing interpreter, how can life emerge without one already in place? The answer is that the interpreter is not a separate thing. It is the physics. The timeline determines which associations hold and which do not. It is only when we abstract away from the timeline and treat symbols as if they have meaning independent of their physical realisation that we create the illusion of a separate interpreter. That illusion is computational dualism (Bennett, 2024a). Dissolving it dissolves the chicken-and-egg problem.

Yes, we cannot see the forest through the trees that are our own representations. We are constrained by our own abstraction layers. But one can reason about what must hold in every possible stack. Change must exist for anything to exist. Change implies states, states imply persistence and destruction, and persistence implies normativity. The rest follows.

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